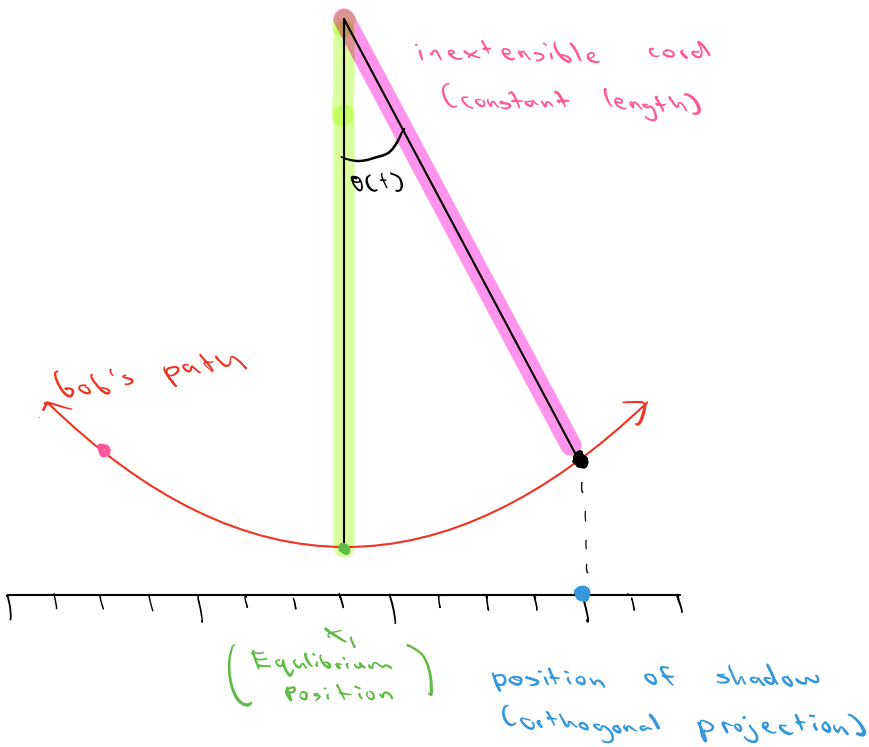


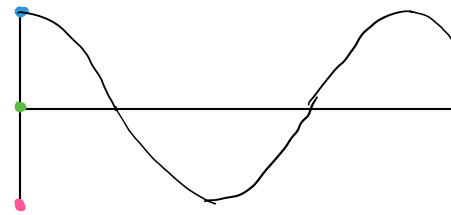
# Motion of Single Pendulum

$x_s(t)$  = position of bob's shadow (orthogonal projection along ruler at any time  $t$ )

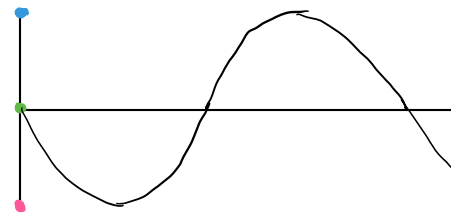
$x_1$  = position of bob's shadow at time  $t$



Displacement  $u(t)$



Velocity  $v(t) = u'(t)$



- This construction helps us study the movement of the pendulum using the movement of its shadow along the ground.
- **linearization**: turning nonlinear motion into linear motion
- For now, we assume we start our pendulum by displacing our bob that represents "small angle" motion.

# Deriving the Equation for a Simple Pendulum

To mathematically model our simple pendulum, we need to derive a differential equation whose solution represents our observed pendulum behavior as a function of time.

$\theta(t)$  = angle between pendulum cord and the vertical reference position

Claim. Any function  $\theta(t)$  that describes our desired angle in pendulum satisfies the differential equation

2nd derivative

$$\ddot{\theta}(t) + \frac{g}{l} \sin(\theta(t)) = 0$$

$g$  = acceleration due to gravity (usually 9.8)

$l$  = measured length of pendulum cord

We can also write this in operator notation as

ordinary differential equation

unknown  $\theta$  desired

single-variable function

$$\theta: D \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

$$\underbrace{\frac{d^2}{dt^2} [\theta(t)]}_{\text{second order term}} + \underbrace{\frac{g}{l} \sin(\theta(t))}_{\text{nonlinear term}} = 0$$

If we could "solve" such equation and produce a function  $\theta(t)$ , we could describe pendulum motion exactly.

Goal: Derive equation starting from "first principles"

$$\ddot{\theta}(t) + \frac{g}{l} \sin(\theta(t)) = 0$$

Recall Newton's Second Law:

$$\vec{\Sigma F} = m \cdot a(t)$$

sum of all forces = mass  $\times$  acceleration

Note that there are only two forces acting on the cord.

$$\vec{\Sigma F} = [\text{tension force}] + [\text{gravity force}]$$

$$\Rightarrow \vec{\Sigma F} = \underbrace{\vec{f}_T + [\vec{f}_g]}_{\text{magnitude \& direction vectors w/ units N}}$$

We know that magnitude

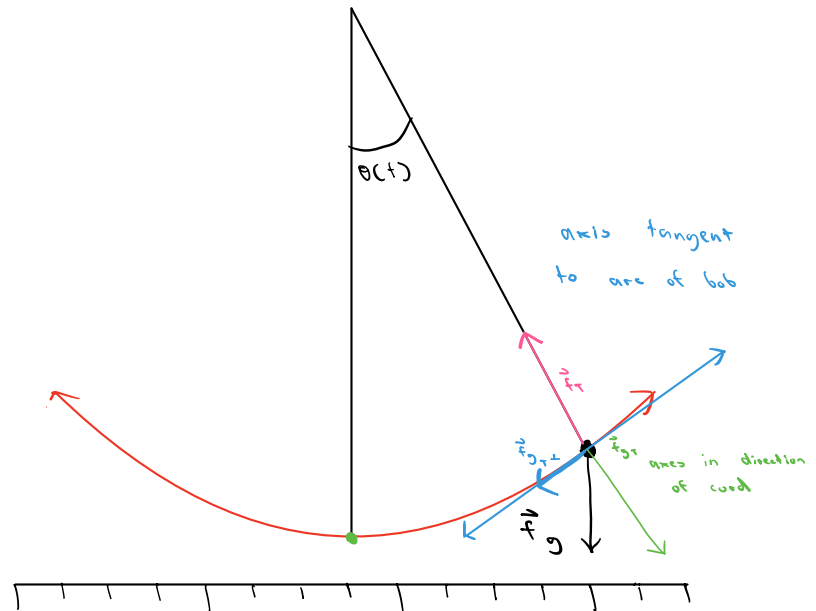
$$\| \vec{f}_g \| = m \cdot g$$

and direction is directly down.

Notice we have

$$\vec{f}_g = \vec{f}_{g_T} + \vec{f}_{g_{T^+}}$$

i.)  $f_{g_T} \perp f_{g_T}^2$



ii.) Since motion of mass inscribes arc-shaped path,  
net force along axis in direction of cord is zero.

$$\Rightarrow \vec{f}_{g_T} + \vec{f}_T = \vec{0}$$

iii.) Net force completely described by  $\vec{f}_{g_{T\perp}}$ :

$$\begin{aligned} \vec{\Sigma F} &= \vec{f}_T + \vec{f}_g \\ &= \vec{f}_T + \vec{f}_{g_T} + \vec{f}_{g_{T\perp}} \\ &= \vec{0} + \vec{f}_{g_{T\perp}} \\ &= \vec{f}_{g_{T\perp}} \end{aligned}$$

iv.) We have some idea about the length of vector  $\vec{f}_{g_{T\perp}}$ :

$$\begin{aligned} \|\vec{f}_{g_{T\perp}}\|_2 &= \|\vec{f}_g\|_2 \cdot |\sin(-\theta(t))| \\ &= m \cdot g \cdot |\sin(-\theta(t))| \end{aligned}$$

How do we assign direction to this force?

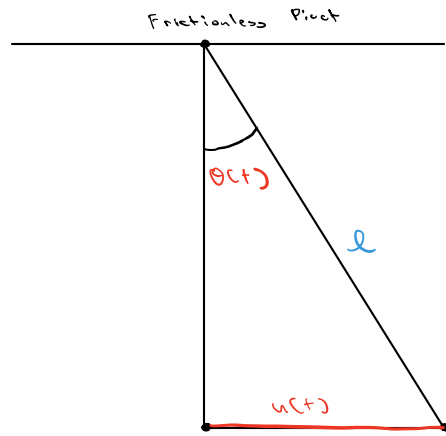
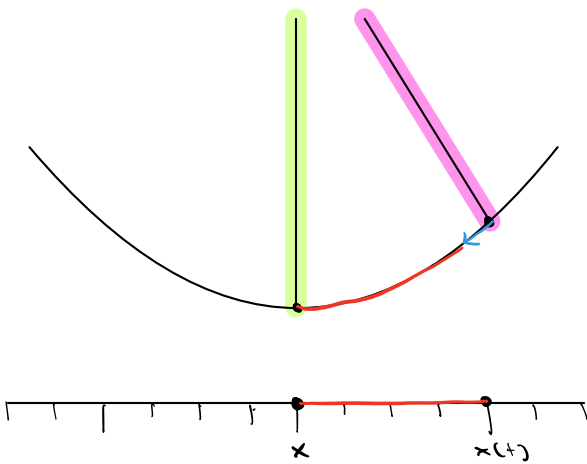
Consider acceleration of mass with respect to time.

$$a(t)$$

Recall that

$$a(t) = \frac{d^2}{dt^2} [u(t)]$$

1h 20 m



Recall:

$$\sin(\theta(t)) = \frac{u(t)}{l} = \frac{\text{opposite}}{\text{hypotenuse}}$$

This implies we can substitute

$$\ddot{\theta}(t) + \frac{g}{l} \sin(\theta(t)) = 0$$

$$\Rightarrow \ddot{\theta}(t) + \frac{g}{l} \cdot \frac{u(t)}{l} = 0$$

However,

$$\sin(\theta(t)) = \frac{u(t)}{l}$$

doesn't immediately suggest how to relate second order terms

$$\ddot{\theta}(t) \text{ and } \ddot{u}(t)$$

